

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)	Minimum answer = force per unit charge (accept force per coulomb)	1			1		
		(ii)	Use of field equation $E = \frac{kQ}{r^2}$ (1) Distance = $0.6\sqrt{2}$ OR $\frac{1.2}{\sqrt{2}}$ OR 0.85 [m] OR equivalent with sin OR cos45 (1) accept $r^2 = 0.72$ [m ²] Penultimate step seen OR 524 5XX seen for field (1) N.B. $k = 9 \times 10^9$ gives exactly 525 000	1	1 1		3	2	
		(iii)	Left (independent) or arrow pointing left (1) $\div \sqrt{2}$ seen or $\times \sin 45$ or $\times \cos 45$ (1) Final answer correct = 1 485 000 [N C ⁻¹] (1)		3		3	2	
	(b)	(i)	Minimum answer – {work done / energy required} per unit charge from infinity (accept {work done / energy required} per coulomb from infinity)	1			1		
		(ii)	[Potential of] +ve and -ve charges cancel (1) Because equidistant OR because the same distance (1) accept P is at the centre OR mathematical: Potential equation used (1) An expression for 4 potentials seen to reduce to zero (k not needed) (1)	1	1		2		
		(iii)	Right and left force confirmed / agreed with (however stated e.g. accelerates right then decelerates is ok) (1) One of the two forces explained (we'll allow 2 nd by implication) e.g. initially repulsed by -ve OR left force later because (closer to) +ve charges (1) $E = 0$ at infinity stated or explained somehow (1)						
						3	3		